

BIOMECHANICS (EPHE 341)

Muscle mechanics

The three properties, and the model they
add up to

By the end of this lecture you should be able to...

- **Explain** how sliding filaments turn overlap into force
- **Describe** the force–length, force–velocity and force–time relationships
- **Assemble** the model $F = F_0 \cdot a \cdot F_L \cdot F_V$
- **Relate** architecture — pennation and cross-sectional area — to force and speed
- **Separate** active from passive tension, and CC from PEC and SEC
- **Read** a work loop, and say what its enclosed area means

Part 01

Where force comes from

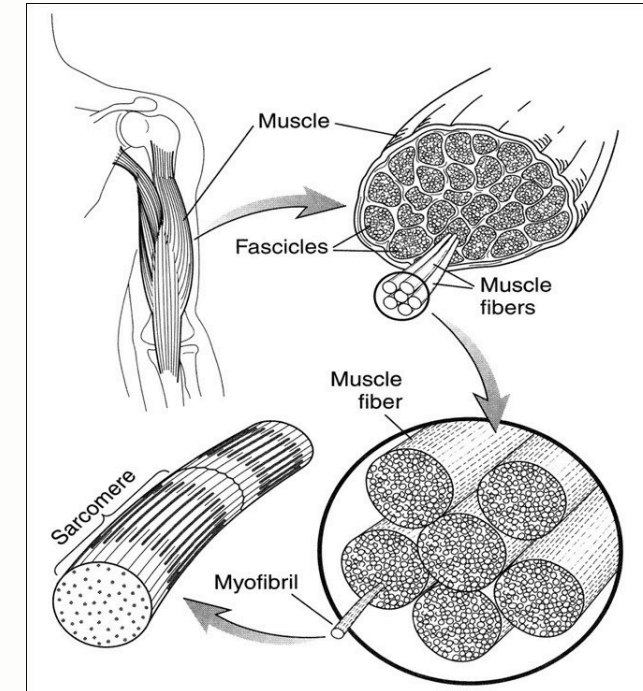
Filaments, architecture, and the moment arm

From muscle to sarcomere

Force is generated at the bottom of this hierarchy

- A muscle is bundles of **fascicles**, each a bundle of **fibres**, each a bundle of **myofibrils**
- A myofibril is a chain of **sarcomeres** laid end to end
- The sarcomere is where the mechanics happens — everything above it is packaging and plumbing
- Each sarcomere is built from two protein filaments: **actin** (thin) and **myosin** (thick)

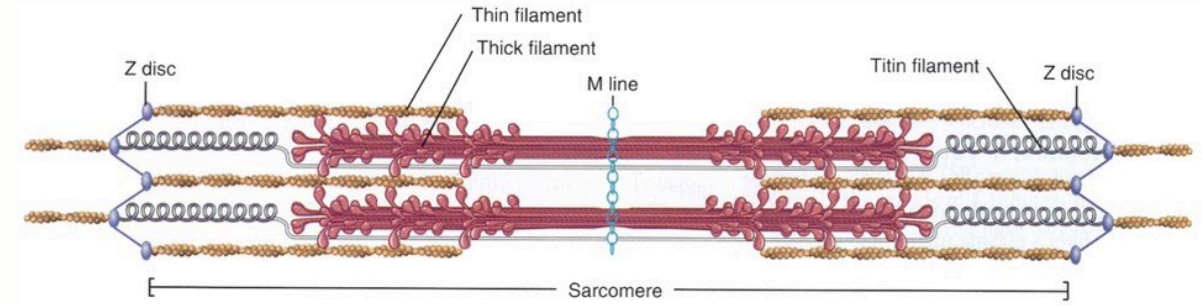
Sarcomeres in **series** give a muscle **range and speed**; sarcomeres in **parallel** give it **force**. Almost every architectural fact in this lecture follows from that one sentence.



The sliding filament

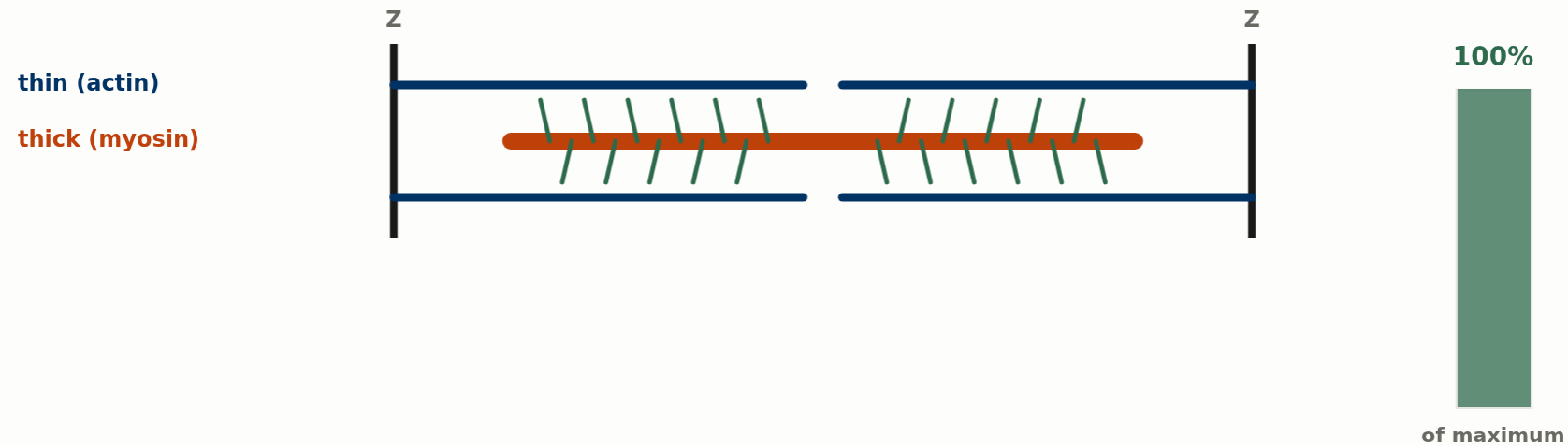
Neither filament changes length — the overlap does

- Force comes from one filament **sliding over** the other, not from either one contracting
- Actin and myosin keep the same length throughout
- What changes is how much of the thick filament has thin filament **alongside it to pull on**
- More overlap means more myosin heads with somewhere to bind — up to the point where the thin filaments start getting in each other's way



The sliding filament

Change the length and watch the overlap — and the force — change



every myosin head has actin to pull on — this is the plateau

sarcomere **2.20 μm** (100% of optimal) · heads with actin to bind **100%** · force **100%** of maximum · neither filament changes length — only the overlap between them does

too short

optimal

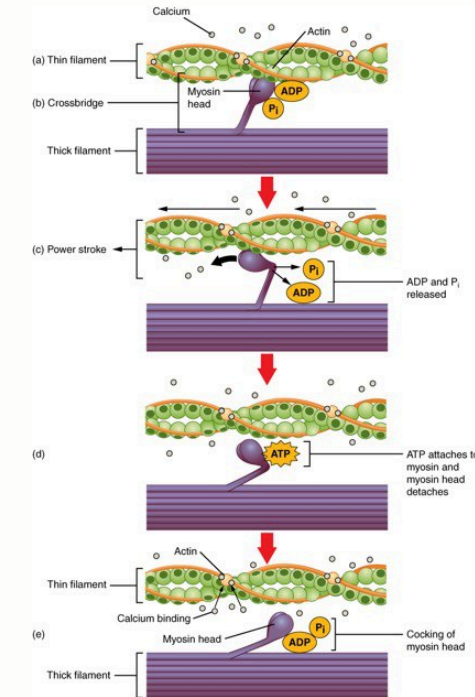
too long

The cross-bridge cycle

One head, one ATP, about ten nanometres

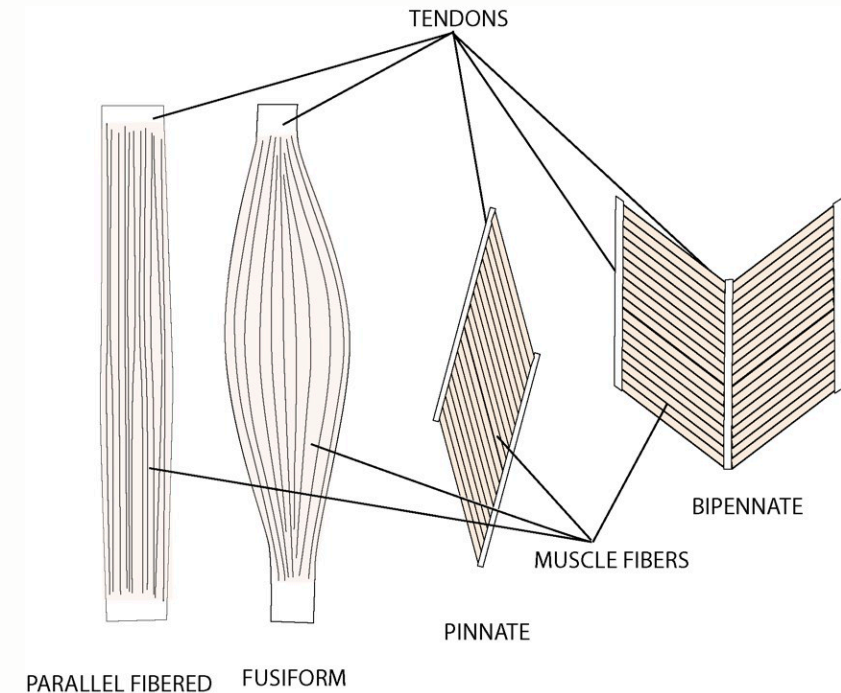
- Side-pieces fixed along the myosin filament **attach** to binding sites on actin, **pull**, then **detach** — over and over
- One cycle moves the filaments about **10 nm** relative to each other
- One cycle produces about **2–10 pN** of force
- One cycle costs the hydrolysis of **one ATP**

A whole muscle's force is a great many of these happening at once and out of step with each other. Heads are always attaching while others are letting go, which is what makes the force smooth rather than ratchety.



The same tissue, arranged four different ways

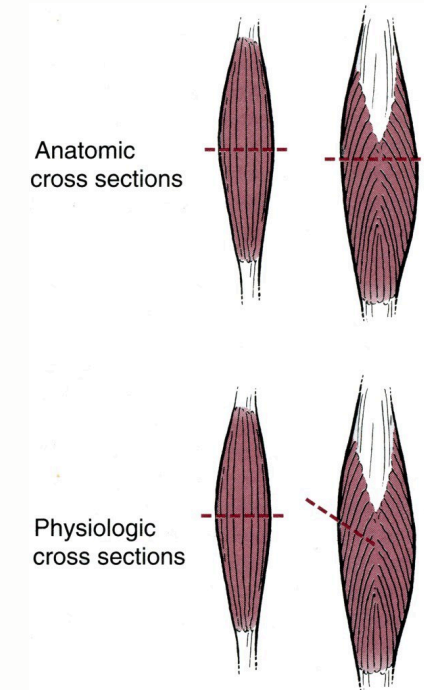
- **Parallel / fusiform** — fibres run the length of the belly. Long fibres, so large range and high shortening speed
- **Pennate / bipennate** — fibres run at an angle to the tendon. Short fibres, so many more of them fit
- Pennation angle is the angle between the **line of action** of the muscle and the direction of its **fibres**
- It changes the direction of every fibre's force vector — and it changes with joint angle, because the muscle keeps its thickness



Physiological cross-sectional area

Count the sarcomeres in parallel — cut perpendicular to the fibres

- The more sarcomeres side by side, the more force the muscle can produce
- You count them by cutting **perpendicular to the fibres**, not perpendicular to the muscle
- For a pennate muscle those are different cuts, and the physiological one is much the larger
- Most muscle produces about **20 N per cm²** of physiological cross-sectional area

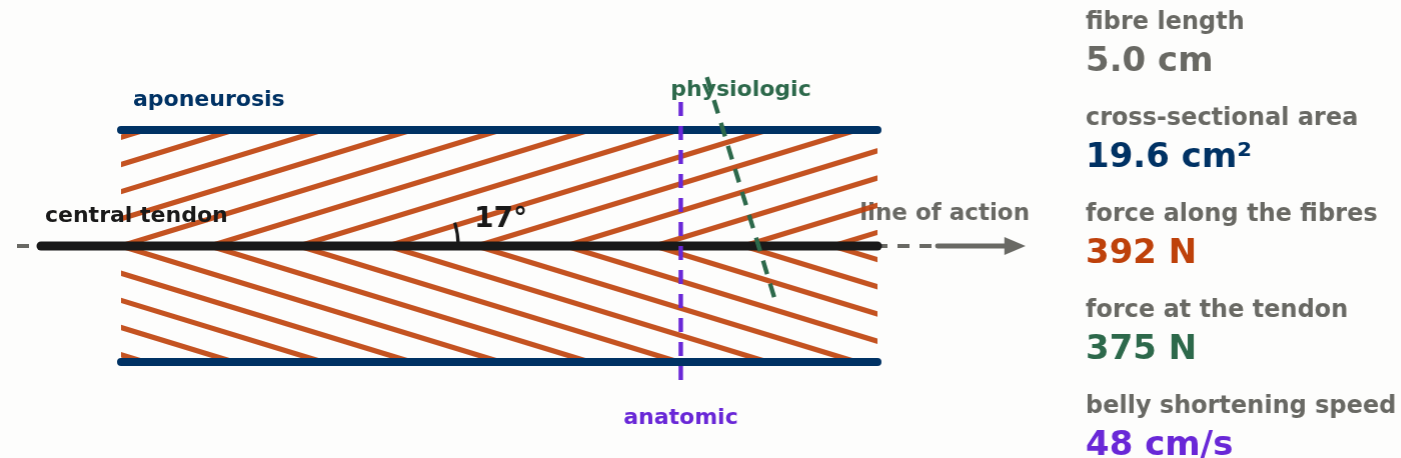


$\text{PCSA} = \text{muscle volume} \div \text{fibre length}$

$F_{\text{fibre}} = \text{PCSA} \times 20 \text{ N/cm}^2$

What pennation buys, and what it costs

The bipennate arrangement from two slides ago, with the numbers attached



against parallel fibres: force $\times 2.3$, speed $\times 0.40$

at 17° · fibres **5.0 cm** · area **19.6 cm²** · fibres pull **392 N**, the tendon gets **375 N** ($\times \cos\theta = 0.956$) · $\cos\theta$ costs a few percent; the shorter fibres buy several times the area — paid for in shortening speed

parallel 0°

gastrocnemius 17°

soleus 25°

deep pennate 35°

Pennation, worked through

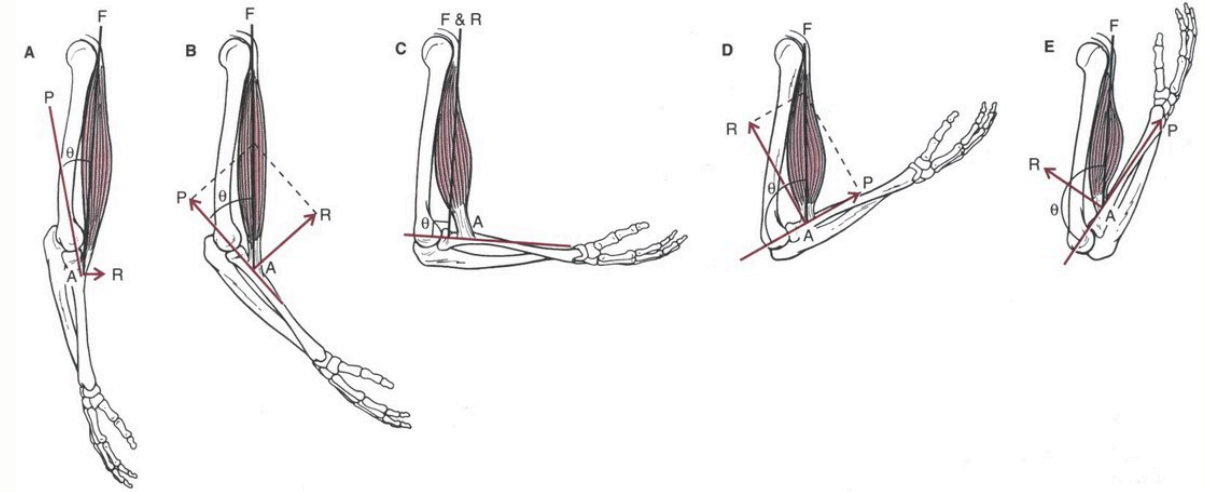
The same muscle volume, arranged two ways

Force is not torque

The moment arm stands between the muscle and the joint

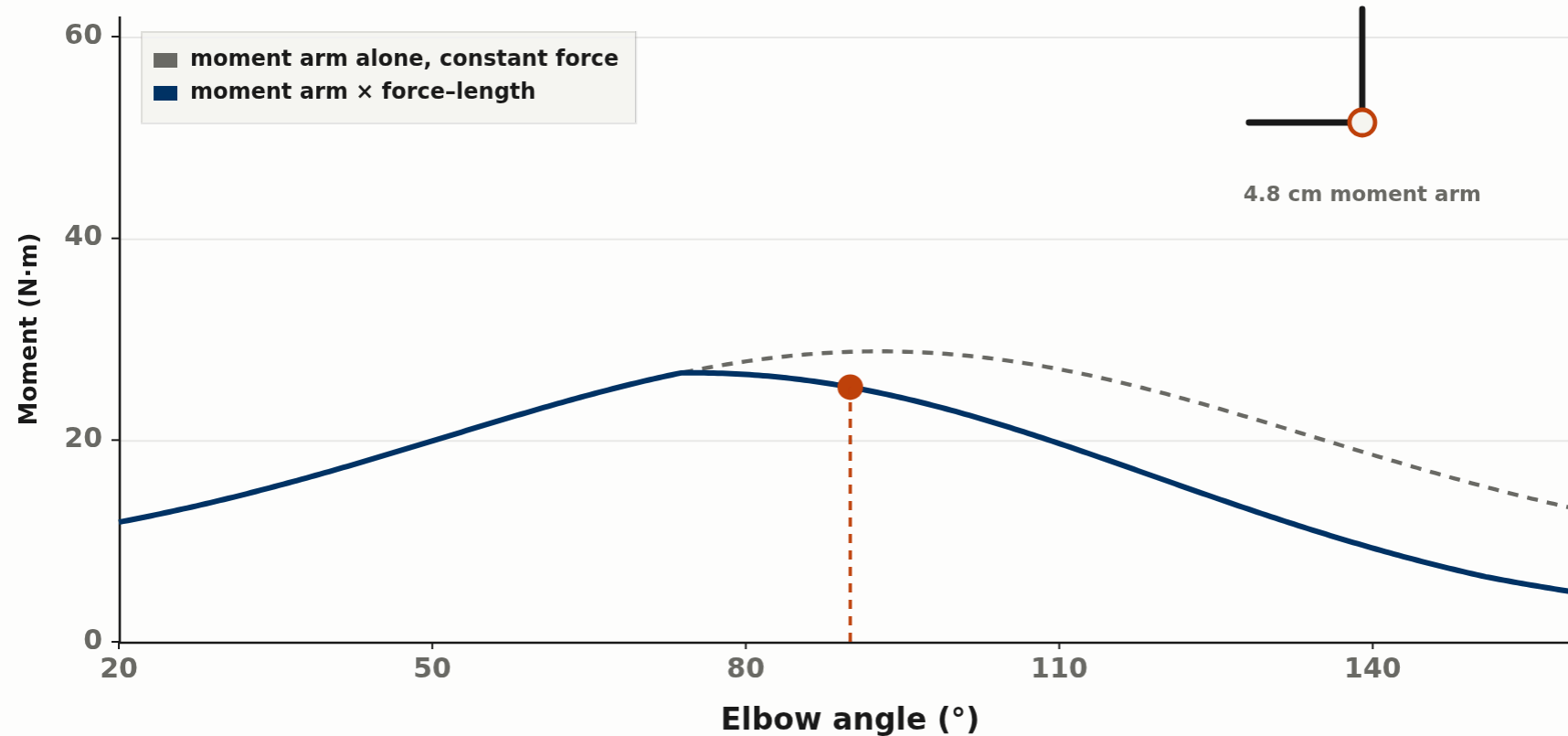
$$T = r(\theta) \cdot F$$

- The moment arm is the perpendicular distance from the muscle's line of action to the joint's axis of rotation
- It **changes with joint angle** — it is a function, not a number
- A bigger moment arm gives more torque for the same muscle force
- But a bigger moment arm also means the muscle must shorten further and faster for the same joint rotation



Force is not torque

Two things change at once as the joint rotates



at **90°** · moment arm **4.8 cm** · force **527 N** · moment **25.2 N·m** · $T = r(\theta) \times F$ — the strength curve you feel is both of these at once, not the muscle alone

Elbow angle



90°

Part 02

The three properties

Force–length, force–velocity, force–time

Three reductionist properties

Muscular force depends on length, on velocity, and on time

FORCE–LENGTH

How much force at this length.
Set by filament overlap, plus passive tissue once the muscle is stretched.

FORCE–VELOCITY

How much force at this shortening or lengthening speed. Set by cross-bridge cycling rates.

FORCE–TIME

How much force *yet*. Force takes time to build and longer to release, because calcium takes time to arrive and to be pumped away.

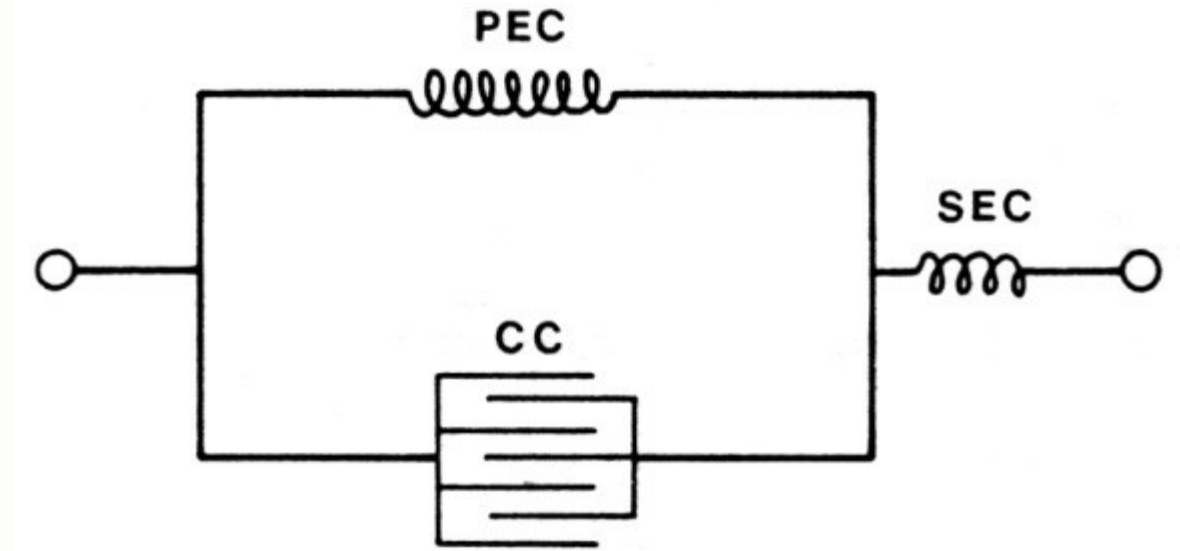
Taken one at a time these are **reductionist** properties — each measured by holding the other two still. Real movement has all three happening at once, and putting them back together is what Part 03 and the Virtual Muscle Lab are for.

A mechanical model of muscle

One active element and two springs

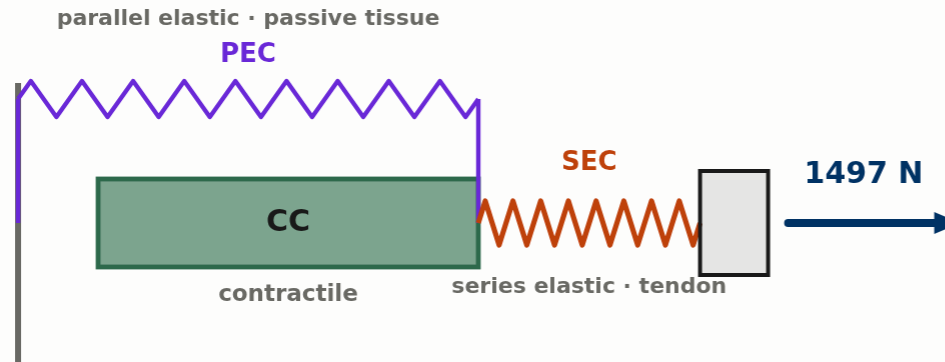
- **Contractile component (CC)** — the cross-bridges themselves, carrying the force–length, force–velocity and activation behaviour
- **Parallel elastic component (PEC)** — connective tissue alongside the fibres, which takes load only when the muscle is stretched
- **Series elastic component (SEC)** — tendon and cross-bridge elasticity, in line between the CC and the bone

Everything the contractile component does reaches the skeleton **through the SEC**. That is why tendon force can lag cross-bridge force, and why a tendon can return energy the muscle never had to produce.



A mechanical model of muscle

Activate it, stretch it, and see where the force comes from



The cross-bridges shorten the CC; the SEC stretches before the bone feels any of it.

activation **80%** · muscle–tendon unit at **100%** of optimal → tendon force **1497 N** · the CC carries the active part, the PEC the passive part, and everything reaches the bone through the SEC

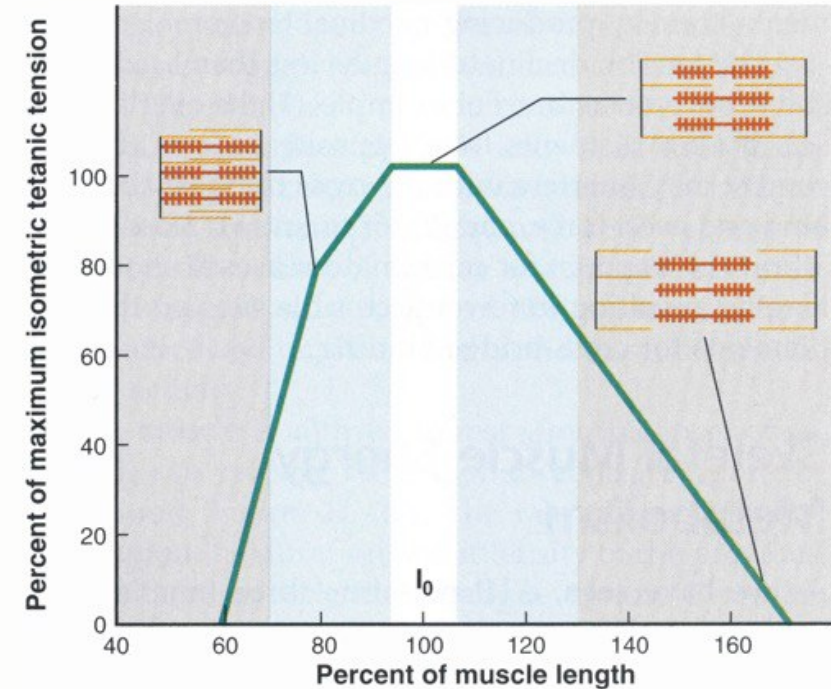
Activation 80%

Stretch of the unit 100%

The overlap story, plotted

- Force depends on how much **actin and myosin overlap**
- Too **short**: thin filaments interfere with each other and the thick filament meets the Z disc
- Too **long**: not enough overlap, so fewer heads can bind
- In between there is an **optimal length**, and a short plateau where every head has somewhere to pull

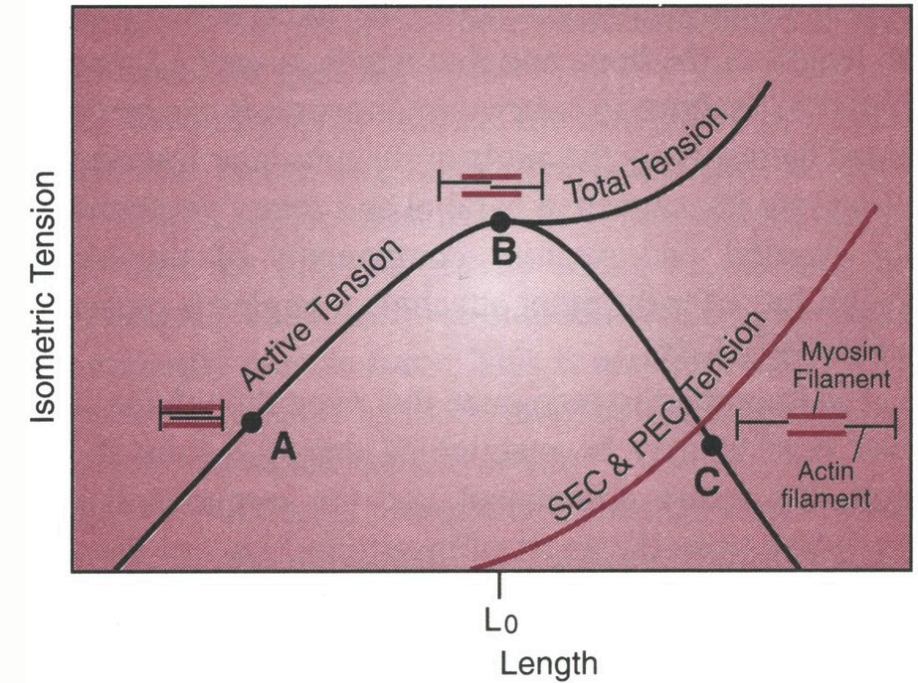
The insets on the figure are the same sarcomere pictures from Part 01. The curve is not an empirical fit — each point on it is a geometry you can draw.



Active and passive tension

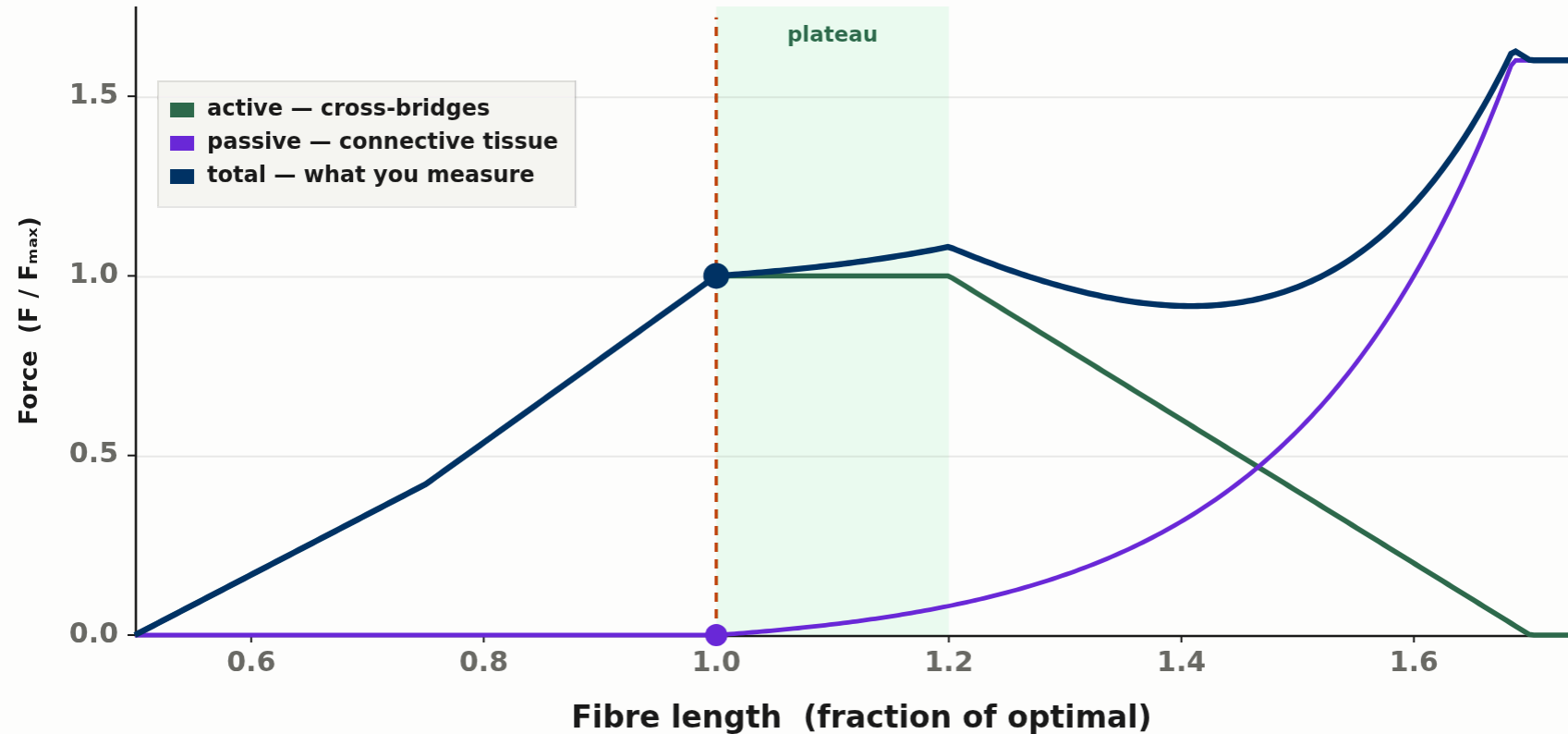
What you measure is the sum of the two

- Tension is generated in **active** tissue — the cross-bridges — and in **passive** tissue, which simply resists being stretched
- Passive tension is **zero** below about optimal length, then rises steeply as connective tissue goes taut
- Add them and you get **total tension**, which is what a dynamometer actually records
- At long lengths most of the force is passive, and the muscle may be producing very little of it



Force-length

Separate the two curves, then add them back



at **100%** of optimal · active **1.00** + passive **0.00** = **1.00** · $F_{\max}$ · near optimal: the plateau, where every head can bind

active + passive

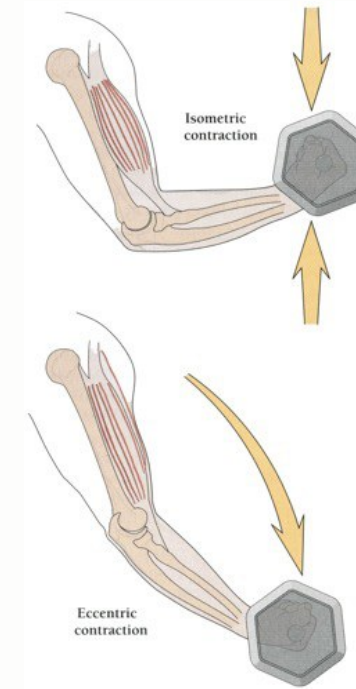
active only

passive only

overlay the Gaussian approximation

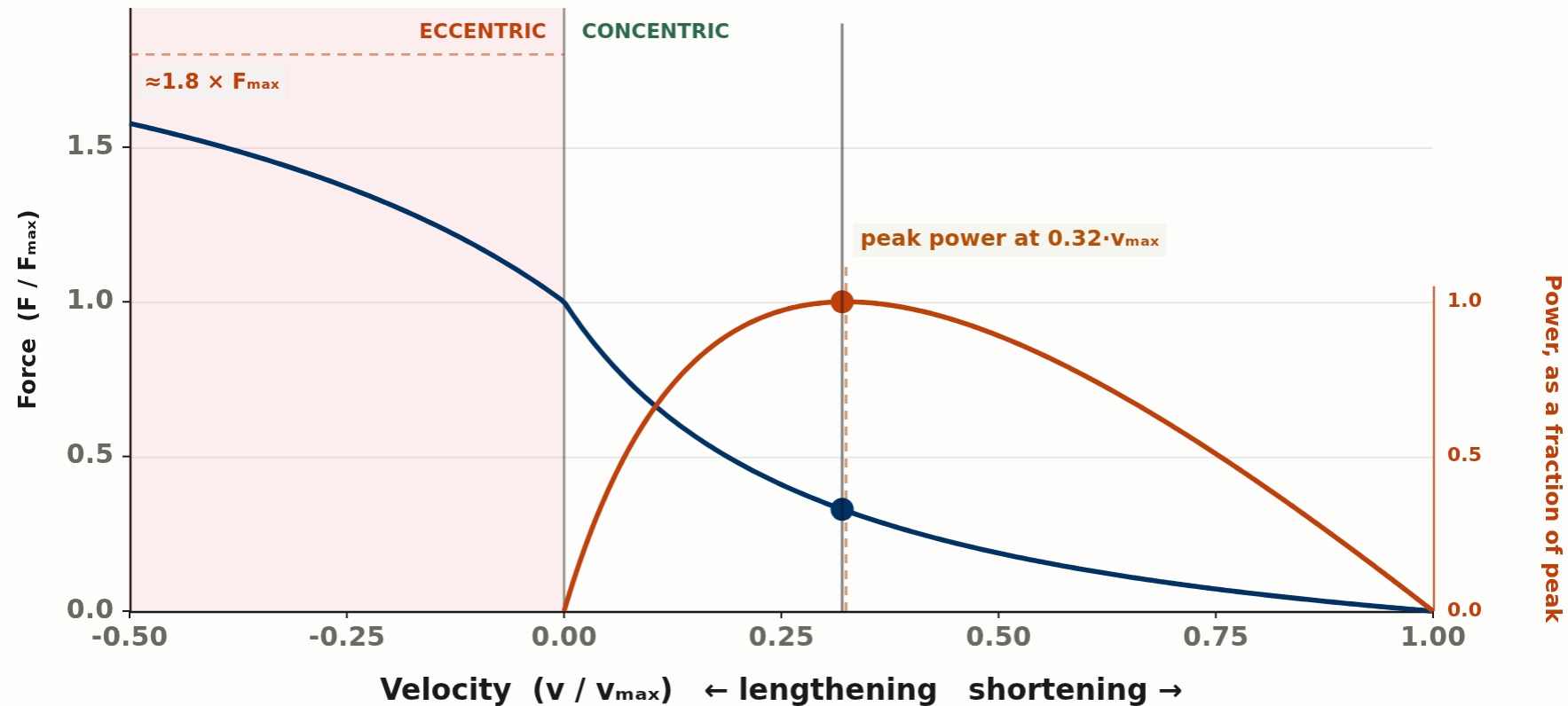
The faster a muscle shortens, the less force it can produce

- **Concentric** — the muscle shortens while producing force. The faster it shortens, the less force, because cross-bridges have less time attached
- **Isometric** — no movement, and maximum force on the shortening side
- **Eccentric** — the muscle lengthens while producing force. Force keeps **climbing**, to roughly **1.8 × maximum**
- That is why you can lower a load you cannot lift, and why eccentric work is such an effective — and such a sore — training stimulus



Force–velocity

Both halves, with the curvature exposed



$v = 0.32 \cdot v_{\max}$ (0.27 m/s) $\rightarrow F = 616 \text{ N} \cdot P = 165 \text{ W}$ produced · curvature $a_e = 0.30$; lower values bend the curve harder and move peak power slower

lengthening

isometric

peak power

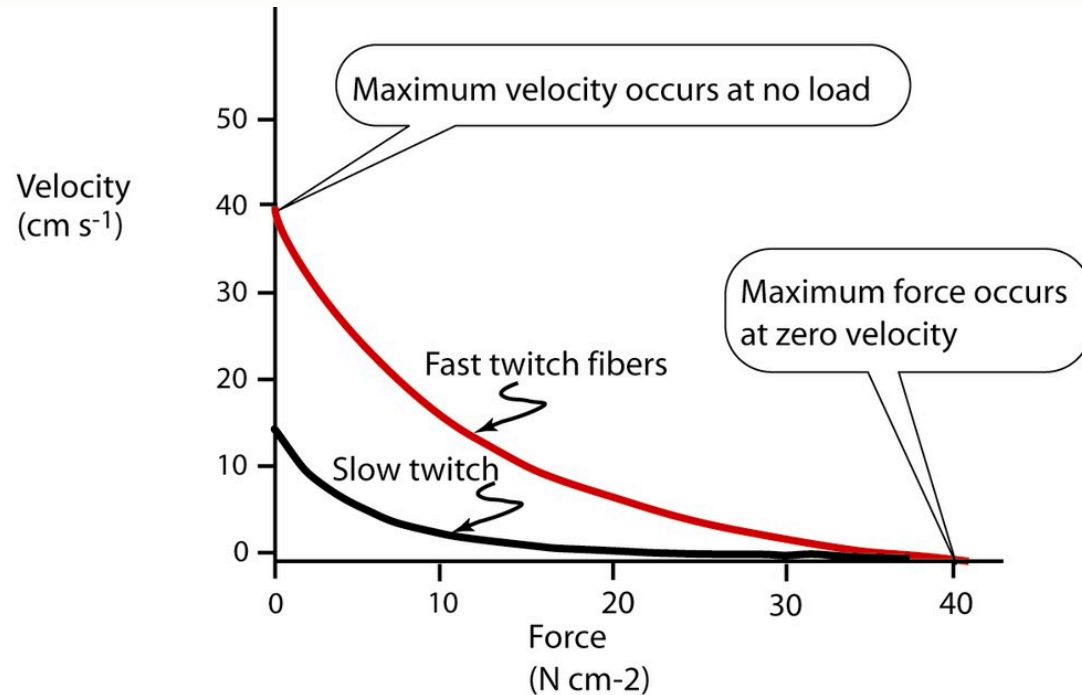
$v_{\max}$

Velocity

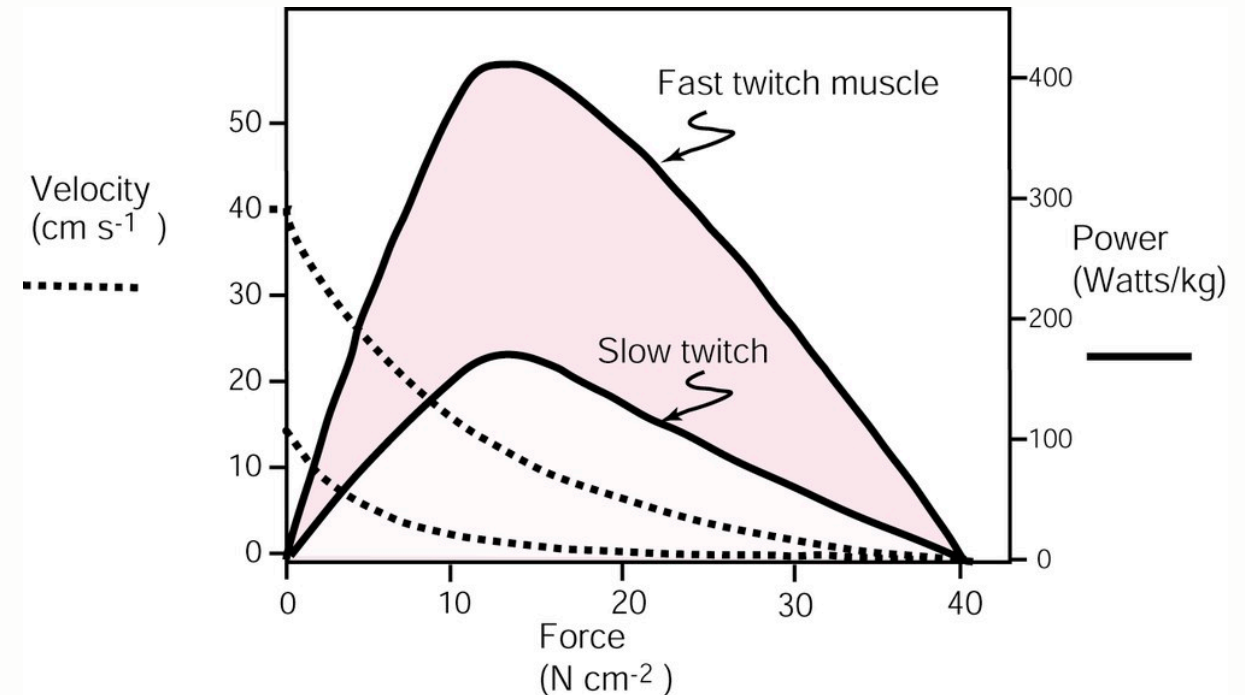
$0.32 \cdot v_{\max}$

Fibre type, and what training can move

The same shape, scaled differently



- Fast-twitch fibres have a much higher $v_{\max}$, so their curve extends much further right
- Peak **power** is therefore far higher in fast-twitch muscle, and occurs at a higher velocity

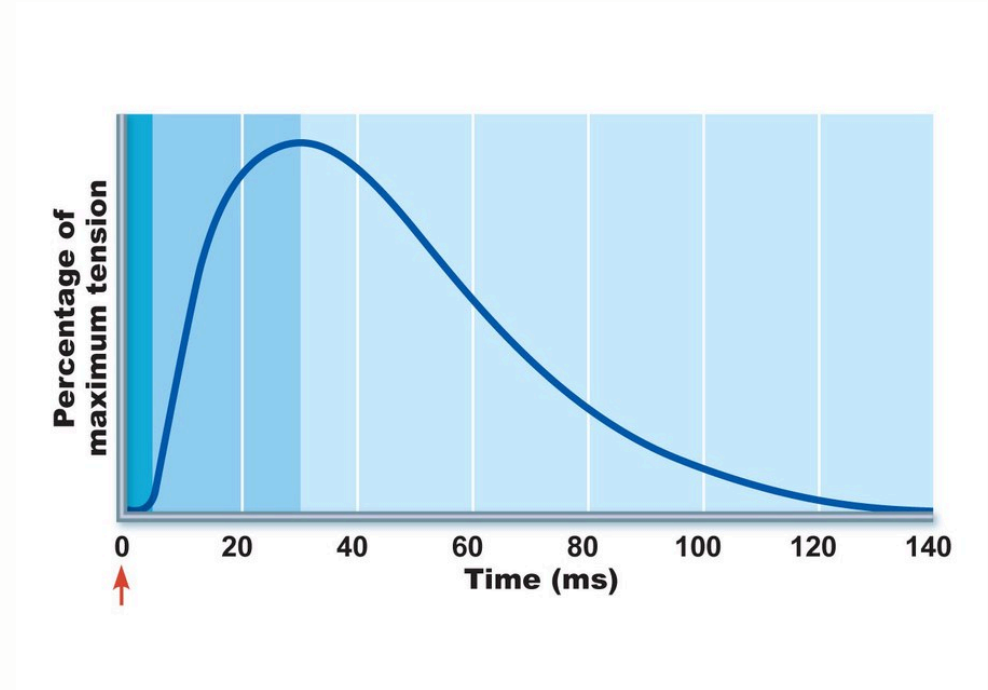


- Both types reach similar force at zero velocity — the difference is **speed**, not strength
- Training shifts the curve; it does not change its shape

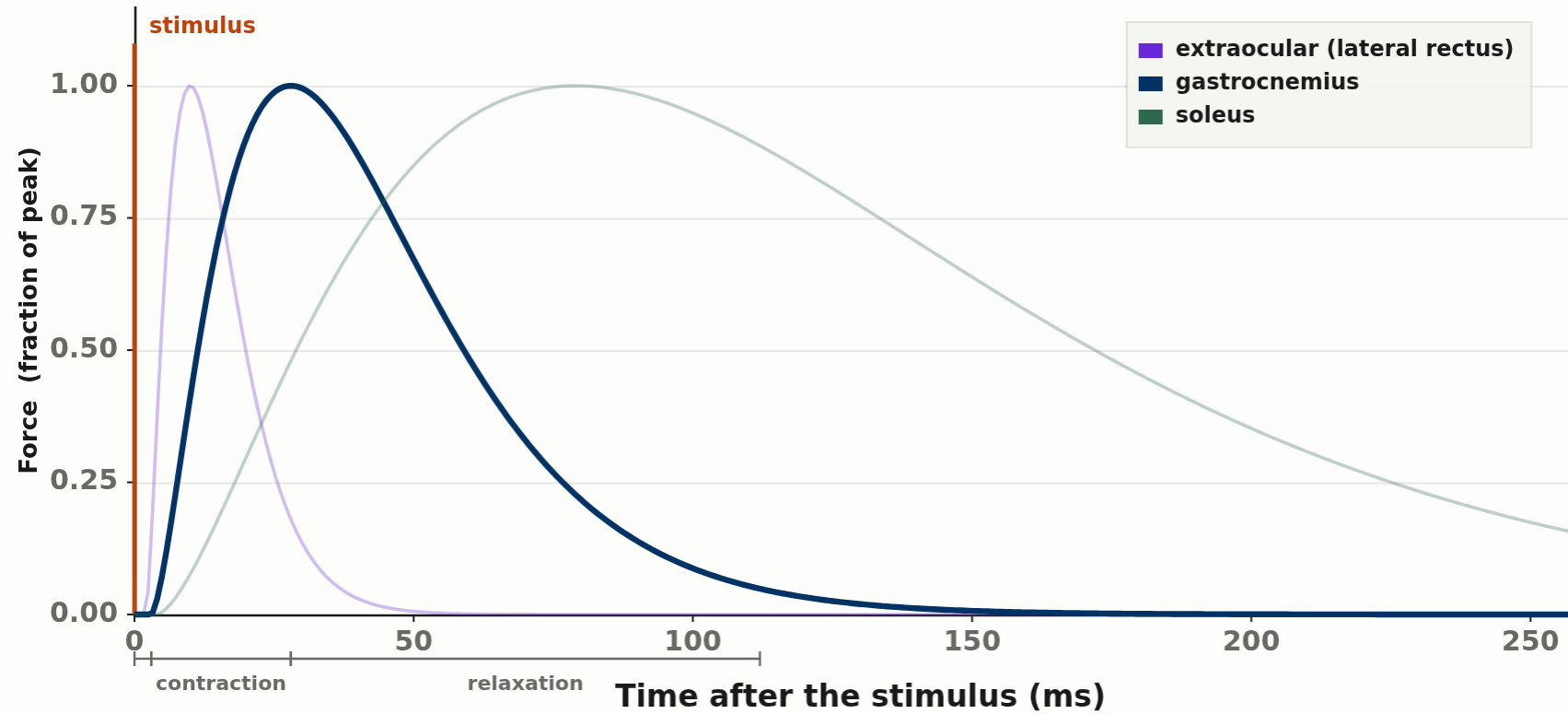
Force does not appear the instant it is asked for

- **Latent period** — the stimulus has arrived and nothing has happened yet
- **Period of contraction** — calcium is spreading and binding, and force is building
- **Period of relaxation** — calcium is being pumped away, and force decays
- Relaxation always takes **longer** than contraction

For a single movement these delays are a curiosity.
For a **cyclic** one — pedalling, running, swimming — they decide when you must switch the muscle on and off, and that is where Part 04 is going.



Three muscles, the same three phases, very different clocks



gastrocnemius · latent **3 ms** · time to peak **25 ms** · half-relaxation **35 ms** · an eye muscle finishes before the soleus has properly started — same mechanism, very different speed

extraocular

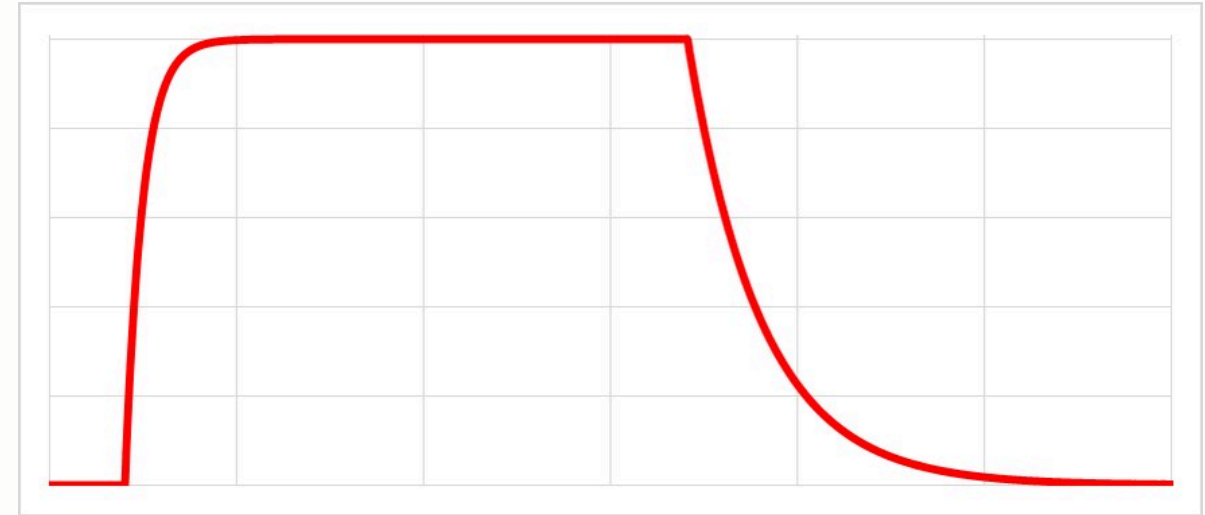
gastrocnemius

soleus

A sustained contraction has a delayed onset and a later offset

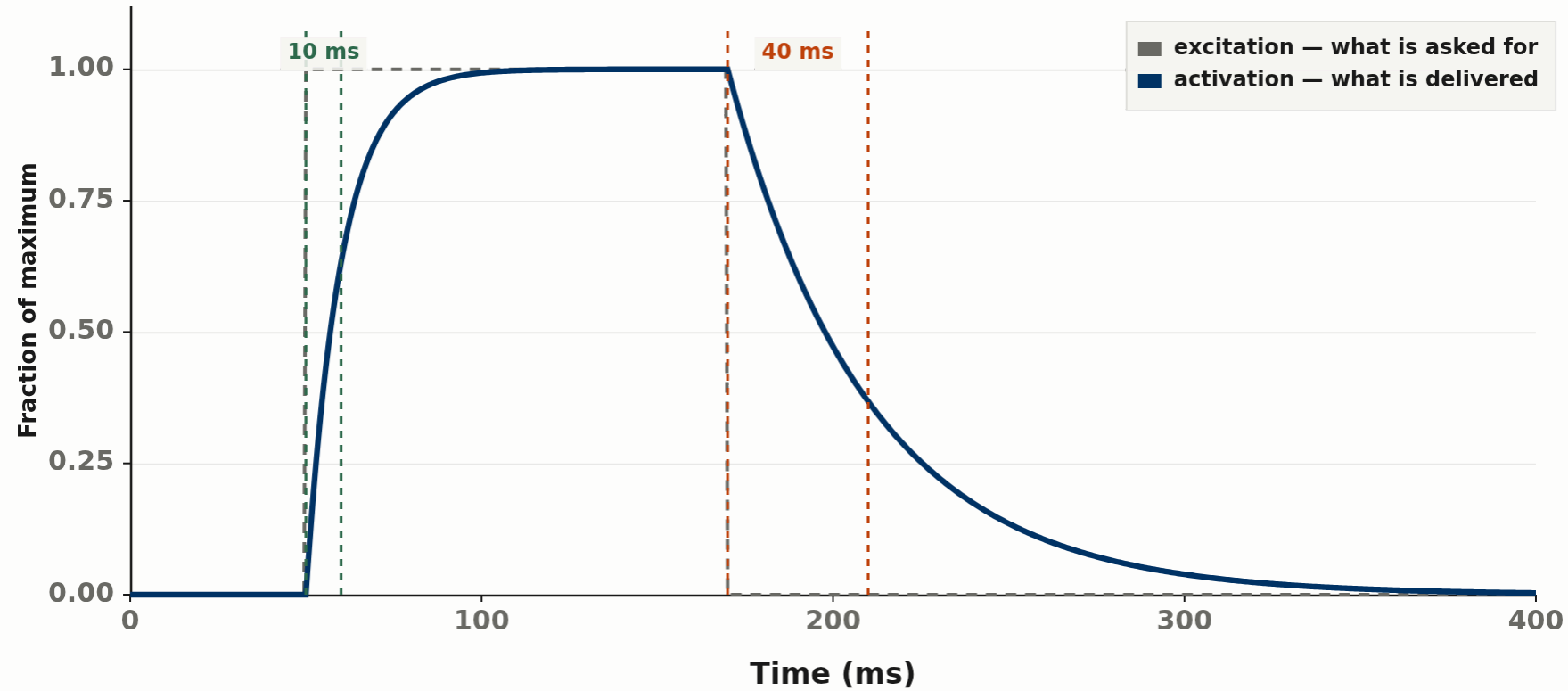
- A held contraction produces an activation that **lags** the signal at both ends
- Rising takes roughly **10 ms**; falling takes roughly **40 ms**
- Both come from the rate at which calcium spreads, binds, and is pumped back
- This is part of — but not the whole of — the **electromechanical delay** between the EMG signal and the force

Those two numbers, 10 ms and 40 ms, are the activation and deactivation time constants in the Virtual Muscle Lab. They are the reason its optimal timing is always *early*.



Activation and deactivation

What is asked for, against what is delivered



τ rise **10 ms** · τ fall **40 ms** · activation reaches 63% after **10 ms** and falls to 37% **40 ms** after the signal stops · turning off takes about four times as long as turning on, and that asymmetry is what makes timing hard

Activation τ

Deactivation τ

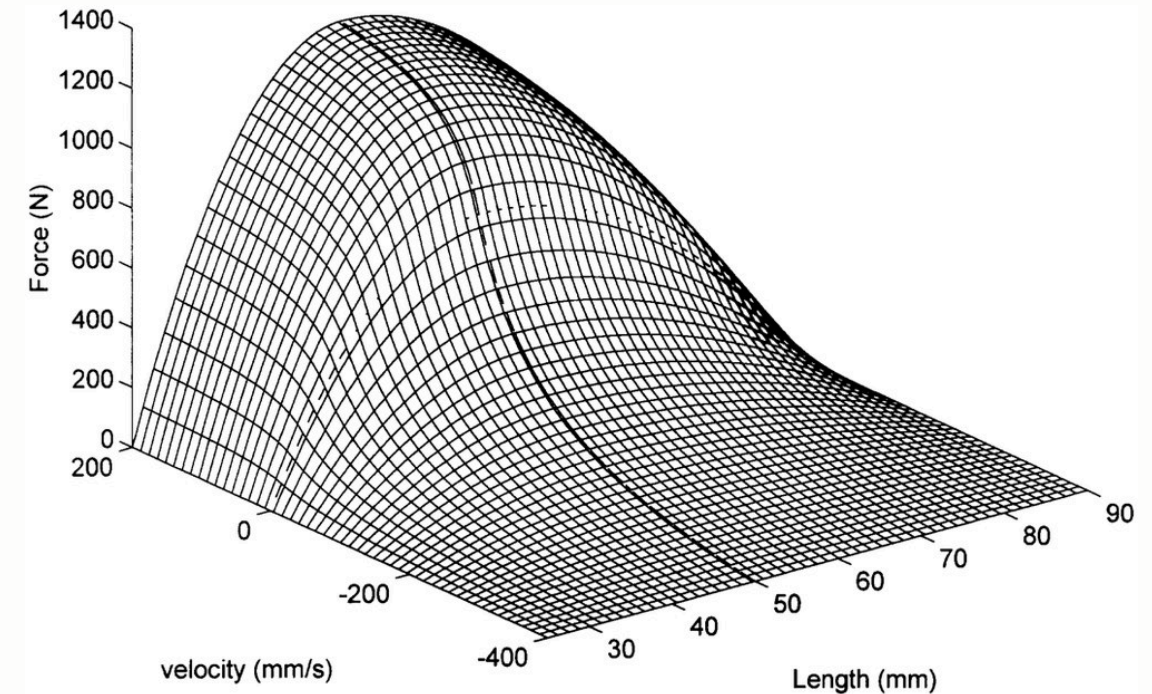
Part 03

Putting them together

One surface, and one equation

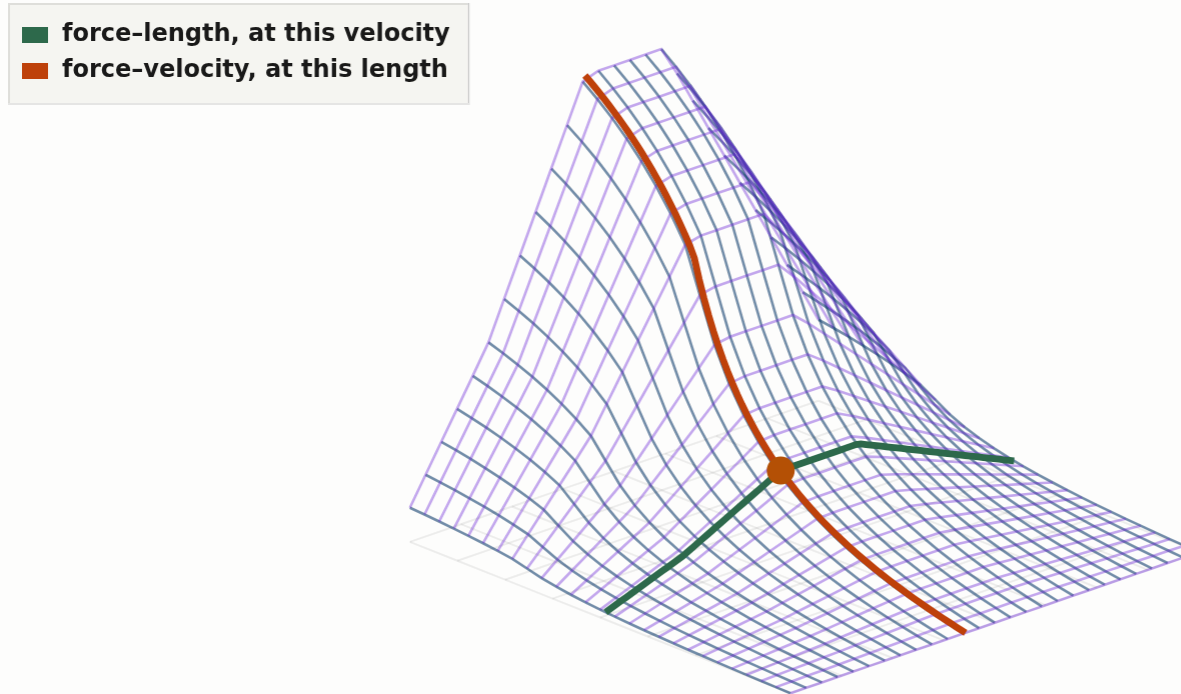
Not two relationships — two cuts through one surface

- Force–length and force–velocity are **acting at the same time**, always
- Plot force against both at once and you get a **surface**
- The force–length curve is a slice across it at one velocity; the force–velocity curve is a slice at one length
- The two factors **multiply**, so being off-optimal in both costs far more than being off in either alone



Force-length-velocity

Drag to turn it, and watch the two familiar curves appear as its edges



drag to turn

length **100%** · velocity **$0.32 \cdot v_{\max}$** → force **33%** of maximum (**616 N**) · $F_l \times F_v$ — the two curves multiply, so being off-optimal in both costs more than either alone

Fibre length



100%

The model equation

Everything so far, in one line

$$F = F_0 \cdot a \cdot F_L(L) \cdot F_V(v)$$

F_0

Maximum isometric force. Set by **PCSA**, pennation and fibre type — the architecture from Part 01.

a

Activation, between 0 and 1. Supplied by the **force–time** behaviour.

$F_L(L)$

The normalised **force–length** factor at the current fibre length.

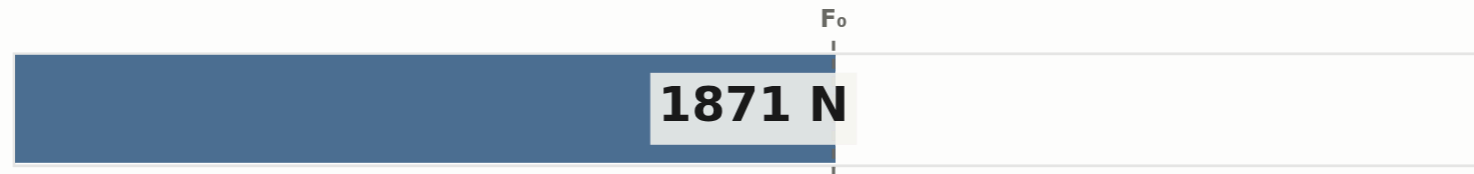
$F_V(v)$

The normalised **force–velocity** factor at the current fibre velocity.

Three of the four terms are fractions between 0 and 1, so they can only take force **away** from F_0 . The one exception is F_V when the muscle is being lengthened, where it rises above 1.

The model equation

Four dials, one product



$$F = F_0 \cdot a \cdot F_1(L) \cdot F_v(v)$$

the same product the Virtual Muscle Lab evaluates at every instant of its cycle

$1871 \times 1.00 \times 1.00 \times 1.00 = \mathbf{1871\ N}$, which is **100%** of maximum isometric · three of the four terms are fractions, so they can only take force away — except F_v when the muscle is being lengthened

Activation a **1.00**

Fibre length L **100%**

From muscle force to joint moment

The full chain, and where each piece came from

$$T = r(\theta) \cdot F(L(\theta), v, a)$$

$$F_0 = F_0(\text{pennation, PCSA, fibre type})$$

- Joint angle θ sets both the moment arm r and the fibre length L — the same variable enters twice
- Joint angular velocity sets the fibre velocity v , through the same geometry
- The nervous system supplies only a
- Everything else is anatomy, and it is fixed for the duration of the movement

This is why a measured joint moment is never a measurement of "muscle strength". It is a measurement of strength, architecture, geometry, length, velocity and activation, all multiplied together.

Part 04

Into the work loop

What happens when the movement is cyclic

A cyclic contraction

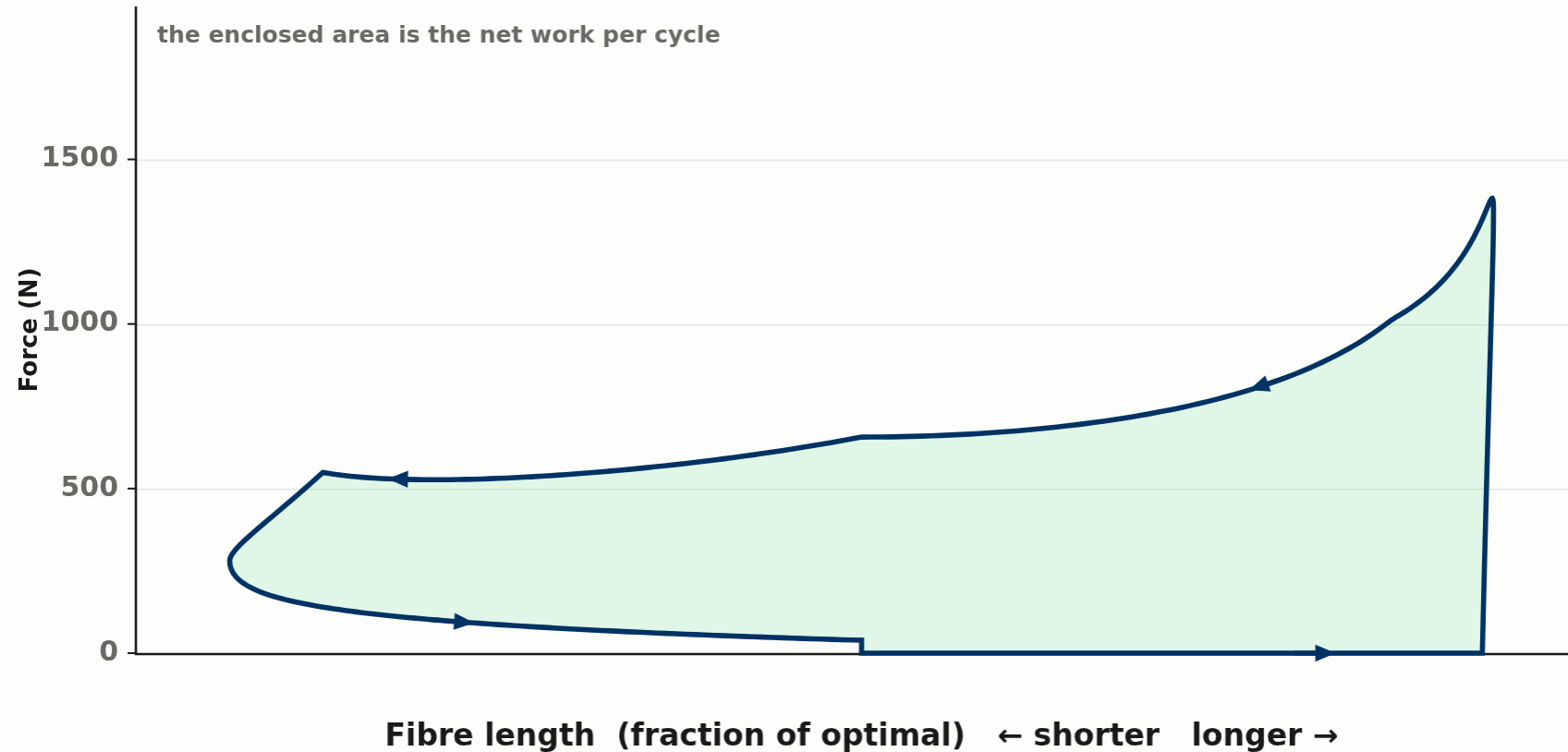
Four things have to happen, in order, every cycle

PHASE	WHAT THE MUSCLE IS DOING	WHICH PROPERTIES GOVERN IT
Activate	Switching on, before it is needed	force–time
Lengthen	Being stretched, under rising load	force–time · force–length · eccentric force–velocity
Shorten	Producing positive work	force–length · force–velocity
Relax	Switching off, before it does harm	force–time

All three properties are in play at every instant, in different proportions. That is why they cannot be understood one at a time, and why the next lecture lets you switch each of them off and watch what is lost.

The work loop

Force against length, round a complete cycle



on at **22%**, off at **66%** · net work **25.14 J** per cycle · mean power **50.3 W** · work × cycle frequency is the power, which is why timing is worth as much as strength

Onset



22%

Next: the Virtual Muscle Lab

The same model, with every property switchable

- The app runs this model through a cycle and reports **work and power**, positive, negative and net
- It computes the same cycle **four times**, turning properties off:

FV, FL and FT	everything, as set
FV and FT	force–length removed
FV and FL	force–time removed — instant activation, ideal timing
F-V only	force–velocity alone

Subtract one column from another and you have measured what that property cost you.

- It will also **optimise** the onset and offset for you, and you can check its answer against the reasoning on the last slide
- Every constant in the sidebar has appeared in this lecture: L_0 , F_0 , $v_{\max}$, curvature, and the two time constants

Come to the next lecture with a prediction: which of the three properties costs the most work at 2 Hz, and does your answer change at 5 Hz?

STRUCTURE

Force comes from filament overlap. Sarcomeres in parallel give force, in series give speed. Pennation trades speed for area, and the moment arm turns force into torque.

THE THREE PROPERTIES

Force–length from overlap plus passive tissue. Force–velocity falling with shortening speed and rising to $1.8 \times F_0$ on lengthening. Force–time, with a 10 ms rise and a 40 ms fall.

THE MODEL

$F = F_0 \cdot a \cdot F_L \cdot F_V$, and $T = r(\theta) \cdot F$. The terms multiply, so deficits compound — and in a cycle, the timing matters as much as any of them.

If you remember one thing: **a muscle's force is never a single number.** It is a product of how much tissue there is, how it is arranged, how long it is, how fast it is moving, and how long ago you asked.

EPHE 341 · BIOMECHANICS

Questions?

Next up: the Virtual Muscle Lab